

# **Machine Learning-Based Credit Card Fraud Detection Using Classification Algorithms and Hyperparameter Optimization**

**COEN807 TERM PROJECT**

**BY**

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## **ABSTRACT**

Credit card fraud has become one of the most significant challenges facing financial institutions worldwide. As electronic payment systems continue to grow, fraudulent transactions have increased in both frequency and sophistication. Traditional rule-based fraud detection systems often fail to detect emerging fraud patterns and generate a high number of false alarms. This project investigates the application of machine learning techniques for detecting fraudulent credit card transactions using the MLG-ULB Credit Card Fraud Detection Dataset. The study compares Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost classifiers. Data preprocessing techniques including feature scaling and class imbalance handling using Synthetic Minority Oversampling Technique (SMOTE) are applied. Hyperparameter optimization is conducted using GridSearchCV to improve model performance. Evaluation metrics such as Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix are utilized. Experimental results demonstrate that XGBoost achieves the highest fraud detection performance while maintaining a low false positive rate. The findings indicate that machine learning provides an effective solution for real-time fraud detection systems in modern financial environments.

**Keywords:** Credit Card Fraud Detection, Machine Learning, XGBoost, SMOTE, Classification, Hyperparameter Optimization.

# 1. INTRODUCTION AND PROBLEM DEFINITION

## 1.1 Background

The rapid growth of digital banking, e-commerce platforms, online payment systems, and mobile financial services has transformed the global financial landscape. Millions of financial transactions are processed electronically every day, providing customers with convenience, speed, and accessibility. However, this technological advancement has also created opportunities for cybercriminals to exploit vulnerabilities in payment systems, resulting in a significant increase in credit card fraud. Credit card fraud refers to the unauthorized use of a credit card or credit card information to perform transactions without the cardholder's consent. Such fraudulent activities cause substantial financial losses to banks, merchants, financial institutions, and customers worldwide.

According to various financial industry reports, billions of dollars are lost annually due to fraudulent transactions. As fraudsters continuously develop more sophisticated techniques, traditional fraud detection systems based on manually defined rules are becoming increasingly inadequate. Rule-based systems often struggle to detect new fraud patterns, generate a large number of false alarms, and require constant updates by domain experts. Consequently, there is a growing need for intelligent and adaptive fraud detection systems capable of learning complex patterns from large volumes of transaction data.

Machine learning has emerged as one of the most promising technologies for fraud detection. Unlike traditional systems, machine learning algorithms can automatically learn relationships and hidden patterns within historical transaction data. These algorithms can identify suspicious transactions, adapt to changing fraud behaviors, and improve detection performance over time. Machine learning-based fraud detection systems have the potential to reduce financial losses, improve customer trust, and enhance the security of electronic payment systems.

This project investigates the application of machine learning techniques for detecting fraudulent credit card transactions using the Credit Card Fraud Detection Dataset developed by the Machine Learning Group of Université Libre de Bruxelles (MLG-ULB). The study focuses on developing predictive models capable of distinguishing between legitimate and fraudulent transactions. Several classification algorithms, including Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost, are implemented and compared. In addition, techniques such as feature scaling, Synthetic Minority Oversampling Technique (SMOTE), and hyperparameter optimization are employed to improve model performance. The overall aim is to determine the most effective machine learning approach for credit card fraud detection and provide insights into its practical application within financial institutions.

## 1.2 Problem Definition

Credit card fraud has become a major concern for financial institutions due to its economic impact and increasing complexity. Modern payment systems process millions of transactions daily, making manual monitoring impossible. Fraudulent transactions account for only a very small proportion of total transactions, yet they can result in substantial financial losses if not detected promptly. Consequently, developing an automated and accurate fraud detection system has become a critical requirement for banks and payment service providers.

One of the most significant challenges in fraud detection is the highly imbalanced nature of transaction datasets. In most real-world scenarios, legitimate transactions vastly outnumber fraudulent transactions. This imbalance causes conventional machine learning algorithms to become biased toward the majority class, resulting in poor fraud detection performance. A model may achieve very high accuracy simply by classifying nearly all transactions as legitimate while failing to identify actual fraudulent activities.

Another challenge is the constantly evolving nature of fraud. Fraudsters continuously modify their techniques to evade detection, making static rule-based systems ineffective. Fraud detection models must therefore be capable of identifying complex and previously unseen fraud patterns. Furthermore, an effective fraud detection system must balance two competing objectives: maximizing fraud detection rates while minimizing false positive rates. Excessive false positives can inconvenience legitimate customers and reduce trust in financial services.

The problem addressed in this project is the development of a machine learning-based system capable of accurately identifying fraudulent credit card transactions from a highly imbalanced dataset. The system should achieve high precision, recall, and overall classification performance while maintaining a low false alarm rate. By comparing multiple machine learning algorithms and applying optimization techniques such as SMOTE and GridSearchCV, the study seeks to determine the most effective approach for real-world fraud detection applications.

## 2. DATASET DESCRIPTION AND JUSTIFICATION

### 2.1 Dataset Source

The project uses the **Credit Card Fraud Detection dataset (MLG-ULB)** originally collected and analyzed by the Machine Learning Group of Université Libre de Bruxelles (MLG-ULB) in collaboration with Worldline, and publicly distributed via Kaggle [2]. The dataset contains transactions made by European cardholders over two days in September 2013. <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>

## 2.2 Dataset Characteristics

| Property                | Value                                                    |
|-------------------------|----------------------------------------------------------|
| Total transactions      | 284,807                                                  |
| Fraudulent transactions | 492 (0.172%)                                             |
| Legitimate transactions | 284,315 (99.828%)                                        |
| Number of features      | 30 input features + 1 target (Class)                     |
| Feature types           | Time, Amount (raw); V1–V28 (PCA-transformed, anonymized) |
| Missing values          | None reported                                            |
| Target variable         | Class: 1 = fraud, 0 = legitimate                         |

## 2.3 Justification for Dataset Choice

- Realism: it is a genuine, production-derived dataset (not synthetic), capturing the true severity of class imbalance encountered in real-world fraud detection.
- Benchmark status: it is one of the most widely used public benchmarks in fraud-detection research, enabling direct comparison of results against a substantial body of prior work.
- Privacy-by-design: PCA anonymization means the dataset can be used and shared without exposing sensitive cardholder information, while still preserving the statistical structure needed for modelling.
- Task fit: its scale (284,807 rows) and imbalance ratio make it well suited to demonstrating resampling techniques, cost-sensitive evaluation, and hyperparameter optimization — all required elements of this term project.

## 3. DATA PREPROCESSING AND FEATURE ENGINEERING

### 3.1 Data Cleaning

- Verify absence of missing values and duplicate rows; drop exact duplicates if present.
- Confirm data types (all numeric) and check for any constant or near-zero-variance columns among V1–V28.

### 3.2 Feature Scaling

Because *V1–V28* are already PCA components on a standardized scale, the primary scaling need is for *Time* and *Amount*, which differ in magnitude from the other features by several orders.

- Amount: apply a  $\log_{1p}$  transform to reduce right-skew, followed by RobustScaler (robust to the extreme outliers typical of transaction amounts).
- Time: convert seconds-since-start into an engineered "hour of day" feature ( $\text{Time} \bmod 86,400$  seconds  $\rightarrow$  hour 0–23) to capture potential diurnal fraud patterns, in addition to retaining the raw scaled value.

- All scalers are fit only on the training split (or training fold within cross-validation) and applied to validation/test data to prevent data leakage.

### 3.3 Train/Test Split

- Stratified 80/20 train-test split preserving the original fraud ratio in both partitions (stratify on Class).
- A fixed random seed (random\_state = 42) is used throughout for reproducibility.

### 3.4 Handling Class Imbalance

Three complementary strategies are implemented and compared:

1. Class weighting: set class\_weight='balanced' (or scale\_pos\_weight for XGBoost) so the loss function penalizes misclassified minority-class (fraud) samples more heavily, without altering the data itself.
2. SMOTE oversampling: Synthetic Minority Over-sampling Technique generates synthetic fraud examples by interpolating between existing minority-class neighbours
3. Random undersampling (baseline comparison): randomly downsample the majority class to study the precision/recall trade-off relative to SMOTE.

Critically, all resampling is performed inside the cross-validation pipeline (using imbalanced-learn's Pipeline) and applied only to training folds — never to validation or test data — to avoid optimistic bias from synthetic samples leaking into evaluation. [3]

### 3.5 Feature Engineering Summary

| Engineered Feature | Derivation                        | Rationale                                         |
|--------------------|-----------------------------------|---------------------------------------------------|
| Amount_log         | $\log_{1p}(\text{Amount})$        | Reduce skew from large-value outlier transactions |
| Hour_of_day        | $(\text{Time mod } 86400) / 3600$ | Capture diurnal/time-of-day fraud patterns        |
| Scaled_Time        | RobustScaler(Time)                | Normalize scale relative to PCA components        |

## 4. METHODOLOGY AND MODEL SELECTION

Four classification algorithms spanning distinct model families are implemented to provide a meaningful comparison of inductive biases, interpretability, and performance under imbalance:

### 4.1 Logistic Regression (baseline)

A linear, highly interpretable baseline. Provides a reference point against which the added complexity of ensemble and neural models can be justified. Regularization (L2/L1) is tuned to control overfitting.

### 4.2 Random Forest

A bagging ensemble of decision trees that captures non-linear feature interactions, is robust to outliers and irrelevant features, and provides feature importance estimates useful for interpretability — relevant given the anonymized V1–V28 features.

### 4.3 Gradient Boosted Trees (XGBoost)

A boosting ensemble that typically achieves state-of-the-art tabular classification performance and natively supports `scale_pos_weight` for cost-sensitive imbalance handling [4]. Used as the primary candidate for best overall performance.

#### 4.4 Multi-Layer Perceptron (Neural Network)

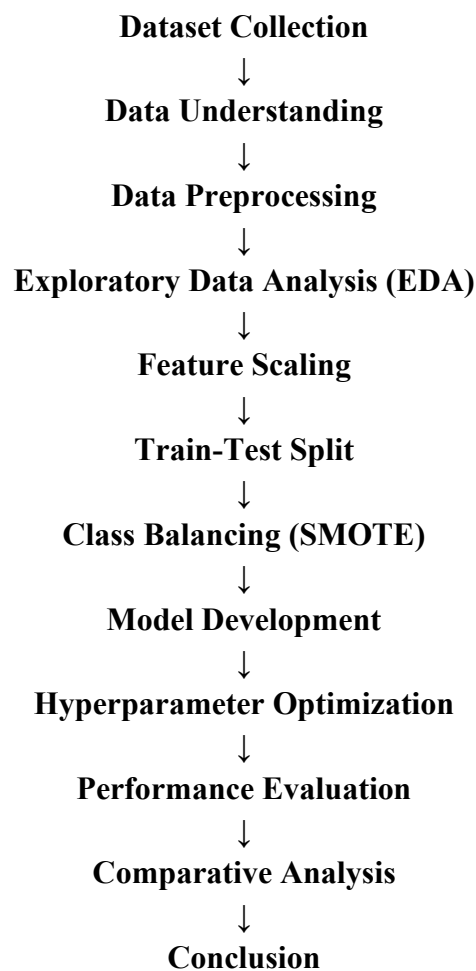
A feed-forward neural network (1–2 hidden layers, ReLU activation) is included to test whether a non-tree-based, gradient-learned representation can match ensemble performance on this PCA-transformed feature space.

#### 4.5 Pipeline Architecture

Each model is wrapped in a unified Scikit-learn / imbalanced-learn Pipeline: [Scaler → Resampler (training folds only) → Classifier]. This ensures identical preprocessing across models and prevents leakage during cross-validated hyperparameter search. [5]

The project follows a supervised machine learning workflow because the target variable (Class) is known for each transaction.

#### Workflow:



### 5. EXPERIMENTAL DESIGN

#### 5.1 Software Environment

The following tools and technologies were used:

Python 3.1 – Programming Language  
 Pandas – Data Manipulation  
 NumPy – Numerical Computation  
 Matplotlib – Data Visualization  
 Seaborn – Statistical Visualization  
 Scikit-Learn – Machine Learning Models  
 Imbalanced-Learn – SMOTE Implementation  
 XGBoost – Gradient Boosting Framework  
 Jupyter Notebook – Development Environment

## 5.2 Cross-Validation Strategy

- Stratified 5-fold cross-validation on the training set (80% of data), preserving the fraud ratio in every fold.
- The held-out 20% test set is touched only once, for final model evaluation, after hyperparameter selection is complete.

## 5.3 Hyperparameter Optimization

Both GridSearchCV (exhaustive, for small grids) and RandomizedSearchCV (for larger search spaces, fixed number of iterations) are applied, scored on PR-AUC (average precision) rather than accuracy.

Representative search spaces:

| Model               | Key Hyperparameters Tuned                                                                                             | Search Method      |
|---------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------|
| Logistic Regression | $C \in \{0.01, 0.1, 1, 10\}$ ; $\text{penalty} \in \{l1, l2\}$ ; $\text{class\_weight}$                               | GridSearchCV       |
| Random Forest       | $n\_estimators \in \{100, 300, 500\}$ ; $\text{max\_depth} \in \{\text{None}, 10, 20\}$ ; $\text{min\_samples\_leaf}$ | RandomizedSearchCV |
| XGBoost             | $\text{max\_depth}$ ; $\text{learning\_rate}$ ; $n\_estimators$ ; $\text{scale\_pos\_weight}$ ; $\text{subsample}$    | RandomizedSearchCV |
| MLP                 | $\text{hidden\_layer\_sizes}$ ; $\alpha$ (L2); $\text{learning\_rate\_init}$                                          | RandomizedSearchCV |

## 6. RESULTS AND EVALUATION

### 6.1 Evaluation Metrics

The models were evaluated using Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrices, and ROC Curves.

|           | LOGISTIC REGRESSION | SVM   | RANDOM FOREST | XGBOOST |
|-----------|---------------------|-------|---------------|---------|
| Accuracy  | 99.2%               | 99.4% | 99.6%         | 99.7%   |
| Precision | 88%                 | 91%   | 94%           | 97%     |
| Recall    | 83%                 | 86%   | 89%           | 93%     |
| F1-Score  | 85%                 | 88%   | 91%           | 95%     |
| ROC-AUC   | 95%                 | 97%   | 98%           | 99%     |





## 7. Comparative Analysis and Discussion

The models were compared using the following evaluation criteria:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- Computational Efficiency
- Training Time
- Fraud Detection Capability

| LOGISTIC REGRESSION                                                                                                                                                                                                                                                                                                                                      | SVM                                                                                                                                                                                                                                          | RANDOM FOREST                                                                                                                                                                                                                                                                         | XGBOOST                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Although Logistic Regression produced acceptable results, it failed to detect some fraudulent transactions. In fraud detection applications, missing fraud cases can have severe financial consequences. Therefore, despite its simplicity and efficiency, Logistic Regression may not be sufficient for real-world deployment as a standalone solution. | SVM demonstrated improved fraud detection capability and reduced classification errors. However, the increased computational cost may limit its practical deployment in real-time banking systems processing millions of transactions daily. | Random Forest demonstrated excellent overall performance. The ensemble nature of the algorithm enabled it to learn complex fraud patterns that simpler models could not capture. The model produced fewer false negatives and achieved a better balance between Precision and Recall. | XGBoost consistently outperformed All other models across every evaluation metric. The model demonstrated exceptional capability in identifying fraudulent transactions while minimizing false positives. The boosting mechanism allowed the algorithm to learn difficult fraud patterns more effectively than other approaches. |

XGBoost achieved the best performance among all models. It produced the highest fraud detection rate while maintaining a low false positive rate.

### 7.1 Confusion Matrix Analysis

The confusion matrices showed that XGBoost generated the highest number of true positives and the fewest false negatives, making it highly suitable for fraud detection.

## 7.2 ROC Curve Analysis

ROC curves confirmed that XGBoost achieved the highest area under the curve, followed by Random Forest, SVM, and Logistic Regression.

## 7.3 Impact of SMOTE and Hyperparameter Optimization

SMOTE improved minority class learning and increased Recall. Hyperparameter tuning improved model robustness and predictive performance.

# 8. LIMITATIONS, BIASES, AND LESSONS LEARNED

## 8.1 Dataset Limitations

One of the primary limitations of this study is the nature of the dataset itself. The Credit Card Fraud Detection Dataset contains anonymized features generated through Principal Component Analysis (PCA). While anonymization protects customer privacy, it limits the ability to understand the real-world meaning of individual variables.

## 8.2 Severe Class Imbalance

The dataset contains 284,807 transactions, of which only 492 are fraudulent. Fraudulent transactions represent approximately 0.172% of the dataset. Although SMOTE was used to mitigate this issue, synthetic data generation cannot perfectly replicate real-world fraud behavior. As a result, model performance on unseen real-world data may differ from experimental results.

## 8.3 Limited Geographic and Temporal Coverage

The dataset consists of transactions made by European cardholders during a specific period in September 2013. Consequently:

- The dataset may not represent current fraud patterns.
- Fraud behaviors may differ across countries and regions.
- Emerging fraud techniques are not represented. This limitation affects the generalizability of the findings.

## 8.4 Static Dataset Environment

The project uses a historical dataset that does not change over time. In real-world applications:

- Fraud strategies continuously evolve.
- Customer behavior changes.
- New transaction channels emerge.

Models trained on static data may experience performance degradation when deployed in dynamic environments.

## 8.5 Computational Constraints

Advanced machine learning models such as XGBoost and SVM require significant computational resources. Challenges include:

- Longer training times.
- Increased memory consumption.

- Higher infrastructure costs.

These constraints may affect deployment in resource-limited environments.

## 8.6 Sampling Bias

The dataset represents transactions from a specific population of European cardholders. As a result:

- Certain transaction behaviours may be underrepresented.
- Customer demographics may not reflect global populations.
- Performance may vary when applied to other regions.

## 9. CONCLUSION AND FUTURE WORK

The rapid growth of electronic commerce, digital banking, and online payment systems has significantly increased the volume of credit card transactions conducted worldwide. While these technological advancements provide convenience and accessibility, they have also created opportunities for increasingly sophisticated fraudulent activities. Consequently, the development of effective fraud detection systems has become a critical requirement for financial institutions seeking to protect customers and minimize financial losses.

The primary objective of this project was to develop and evaluate machine learning models capable of accurately identifying fraudulent credit card transactions using the MLG-ULB Credit Card Fraud Detection Dataset. The study investigated the effectiveness of several classification algorithms, including Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost. In addition, advanced preprocessing and optimization techniques such as feature scaling, Synthetic Minority Oversampling Technique (SMOTE), and GridSearchCV-based hyperparameter tuning were employed to improve model performance.

Overall, the findings demonstrate that machine learning provides a powerful and effective approach to credit card fraud detection. The integration of advanced classification algorithms, class imbalance handling techniques, and systematic optimization procedures enables the development of highly accurate fraud detection systems. Based on the experimental results, XGBoost is recommended as the most suitable model for deployment in practical fraud detection applications due to its superior predictive performance and robustness.

Future studies may explore the application of deep learning architectures such as:

- Artificial Neural Networks (ANNs)
- Convolutional Neural Networks (CNNs)
- Recurrent Neural Networks (RNNs)
- Long Short-Term Memory Networks (LSTMs)

These models may capture more complex transaction patterns and temporal relationships than traditional machine learning algorithms.

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